PROJECT REPORT

HOSPITAL MANAGEMENT SYSTEM

1. Introduction

SYSTEM OVERVIEW

A console-based application developed in C, designed to automate and streamline hospital operations.

CORE PURPOSE

Digitizing patient records and doctor schedules to reduce manual errors and improve administrative efficiency.

2. Core System Objectives

Centralized Platform: Managing patient admissions and discharges efficiently.

Organized Database: Maintaining records of medical staff and specializations.

Automated Billing: Ensuring accuracy in financial transactions and billing.

Data Persistence: Using robust file handling for secure data storage.

3. Features & Functionalities

PATIENT ADMISSION

Captures **ID, name, age, and gender**. Manages ward and bed allocations to prevent double-booking.

DOCTOR MANAGEMENT

Registers new medical staff and tracks their **specializations and contact details** in a structured database.

DOCTOR ASSIGNMENT

Facilitates the **linking of patients** to specific doctors, ensuring specialized care for every admitted individual.

BILLING SYSTEM

Automated calculation of **room charges, medicine, and fees** with built-in logic for service charges and discounts.

4. Technical Specifications

COMPONENT	SPECIFICATION
Programming Language	C (Standard C11)
Data Structures	Structs, Arrays, Pointers
Storage System	Binary Files (.dat)
User Interface	Command Line Interface (CLI)
Compiler	GCC (GNU Compiler Collection)

5. System Modules Breakdown

MANAGEMENT MODULE

Handles the **core logic** for admitting patients and registering new medical staff into the system.

INFORMATION MODULE

Responsible for **retrieving and displaying** comprehensive lists of patients and doctors.

FINANCIAL MODULE

Manages **billing algorithms**, including stay duration, medicine costs, and automated discount logic.

PERSISTENCE MODULE

Manages **File I/O operations**, ensuring data is securely saved to and loaded from binary files.

6. Conclusion & Future Scope

CONCLUSION

The **Hospital Management System** successfully addresses the fundamental needs of healthcare administration, providing a reliable and efficient tool for data management and operational workflows.

FUTURE SCOPE

Future enhancements will focus on implementing a **Graphical User Interface (GUI)**, integrating real-time databases, and adding **networking capabilities** for multi-departmental access.

END OF REPORT